

Frequency Domain and Transforms

Problem Sheet problems completed already.

Image enhancement

1. Spatial Domain Filtering
 - a. Low-pass filtering using box filter
 - b. Low-pass filtering using gaussian filter
 - c. High-pass filtering
 - d. Unsharp masking
 - e. High-boost filter
2. Transform domain-based image enhancement
 - a. Generation of low-pass filter mask
 - i. Ideal low-pass filter
 - ii. Butterworth filter
 - iii. Gaussian filter
 - b. Generation of high-pass filter mask
 - i. Ideal high-pass filter
 - ii. Butterworth filter
 - iii. Gaussian filter
 - c. High-frequency emphasis filter in frequency domain
 - d. Ideal band-pass filter in the frequency domain
 - e. Ideal band-reject filter in the frequency domain
3. Homomorphic filter

Image Denoising and Image Restoration

1. Noises (Gaussian, Uniform, Impulse, Speckle, Poisson, Rayleigh, Periodic)
2. Filters for Noise Minimization
 - a. Mean filter
 - b. Averaging and weighted averaging
3. Non-linear spatial domain filter
 - a. Alpha trimmed mean filter
 - b. Minimum filter
 - c. Maximum
 - d. Median
4. Anisotropic diffusion filter
5. Fourier transform-based image denoising
6. Image Restoration
 - a. Point spread function
 - i. Out-of-focus blur
 - ii. Atmospheric turbulence blur
 - iii. Motion blur
 - b. Inverse filter
 - c. Pseudo-inverse filter
 - d. Wiener filter